

SOUTH ARKANSAS, AR, USA

WMO: 723419

Lat: 33.221N

Lon: 92.814W

Elev: 77

StdP: 100.41

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93992

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-3.7	-12.8	1.3	-1.8	-10.0	1.6	0.0	8.6	15.5	8.0	14.3	1.6	20	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	37.0	24.8	35.4	24.8	34.2	24.8	26.8	32.5	26.3	32.2	25.8	31.5	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.7	28.6	24.9	20.1	28.3	24.3	19.4	28.1	84.1	32.6	81.9	32.1	80.0	31.7	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.8	5.7	-9.1	38.9	2.3	2.7	-10.8	40.9	-12.2	42.5	-13.5	44.0	-15.1	45.9		
			-9.9	28.0	2.2	1.2	-11.5	28.9	-12.8	29.7	-14.1	30.4	-15.7	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	7.1	9.1	13.3	17.7	22.2	26.1	27.8	27.7	24.3	18.0	12.2	8.2
	DBStd	8.59	5.76	5.71	5.24	4.17	3.42	2.29	2.09	2.39	3.36	4.50	4.95	5.43
	HDD10.0	376	128	79	27	2	0	0	0	0	0	2	33	104
	HDD18.3	1426	351	262	168	60	10	0	0	0	3	62	192	318
	CDD10.0	3242	38	53	131	234	377	483	552	549	428	251	99	47
	CDD18.3	1251	2	3	13	42	129	233	293	291	181	53	8	3
	CDH23.3	12747	4	19	117	388	1153	2363	3202	3201	1734	509	48	8
	CDH26.7	5948	0	1	17	75	396	1116	1654	1716	810	159	4	0
Wind	WSAvg	2.4	2.7	2.9	3.0	3.0	2.6	2.2	1.9	1.9	1.9	2.0	2.3	2.5
Precipitation	PrecAvg	1223	119	95	127	122	139	78	111	83	96	81	91	108
	PrecMax	1867	245	170	239	248	333	167	237	200	215	228	242	236
	PrecMin	891	36	20	42	12	13	4	14	5	12	0	30	16
	PrecStd	318	64	42	57	75	80	54	57	53	61	67	55	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	25.9	28.6	30.4	32.9	36.3	38.0	39.3	37.0	32.3	27.3	24.6
		MCWB	17.7	18.3	18.7	21.4	24.0	24.4	25.0	24.4	23.6	21.7	20.2	19.2
	2%	DB	21.7	23.0	26.3	28.7	31.7	34.7	36.3	37.4	34.8	30.2	24.6	22.1
		MCWB	16.7	17.2	18.3	20.1	23.2	24.7	25.2	24.8	23.5	21.6	18.6	18.0
	5%	DB	19.5	21.1	24.3	27.4	30.5	33.3	35.0	35.7	33.0	28.5	22.7	19.9
		MCWB	15.9	16.5	17.1	19.6	22.8	24.6	25.3	24.8	23.4	20.4	17.9	17.1
	10%	DB	17.3	18.9	22.3	25.8	29.0	32.3	33.8	34.2	31.4	26.4	20.9	17.7
		MCWB	14.2	14.9	16.2	18.6	22.2	24.4	25.3	24.9	23.1	19.7	17.4	14.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	20.3	20.8	23.4	25.8	26.9	27.7	27.4	26.1	25.0	22.2	20.5
		MCDB	22.4	23.1	25.0	27.7	30.9	32.4	32.7	32.8	31.6	29.3	24.2	22.7
	2%	WB	18.2	19.0	19.9	22.0	24.7	26.2	26.9	26.8	25.4	23.7	20.5	19.2
		MCDB	20.6	22.0	24.2	26.1	29.3	31.7	32.6	32.5	31.0	27.6	22.9	21.3
	5%	WB	16.9	17.6	18.8	21.3	24.0	25.7	26.4	26.2	24.9	22.6	19.4	17.7
		MCDB	18.8	20.1	22.6	25.1	28.5	31.0	32.4	32.2	30.2	25.8	21.8	19.6
	10%	WB	14.6	15.8	17.6	20.3	23.2	25.2	25.9	25.6	24.2	21.4	17.8	15.5
		MCDB	16.8	18.3	21.1	24.1	27.5	30.4	31.7	31.6	29.2	24.4	20.2	17.4
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.8	12.6	12.9	11.6	11.3	11.3	12.1	12.7	13.3	12.4	11.3
		MCDBR	13.6	14.1	14.3	13.6	12.6	12.7	13.0	14.4	14.4	15.1	14.2	12.9
		MCWBR	9.4	9.0	7.3	6.3	5.2	4.5	4.1	4.1	4.9	7.1	8.8	9.0
	5% WB	MCDBR	10.7	11.7	11.9	10.9	10.7	11.0	11.5	11.8	11.7	11.2	11.3	10.9
MCWBR		8.8	8.8	7.4	6.3	5.0	4.4	4.3	4.3	4.4	6.5	8.6	8.6	
Clear-Sky Solar Irradiance	taub	0.327	0.337	0.364	0.395	0.435	0.456	0.483	0.476	0.419	0.363	0.345	0.326	
	taud	2.514	2.484	2.420	2.327	2.274	2.259	2.208	2.222	2.357	2.484	2.488	2.522	
	Ebn at Noon	890	917	913	894	857	835	810	810	846	874	860	870	
	Edh at Noon	85	96	111	127	135	137	144	140	117	95	86	80	
All-Sky Solar Radiation	RadAvg	2.61	2.98	4.24	5.28	5.83	6.35	6.22	5.80	4.97	4.00	2.87	2.29	
	RadStd	0.25	0.50	0.36	0.35	0.44	0.50	0.41	0.49	0.45	0.54	0.33	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page